Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Ctrl + Sleep
Project / Solution Name	Quote bot AI
Name (Leader)	Chin Xian Jing
Email & Phone (Leader)	Xianjing0702@gmail.com
University Affiliation(s)	University of Nottingham Malaysia
Live Prototype URL	https://sdc-hackathon-jovilim67.replit.app Staff password : aaaaaaaaaaaaaa
Code Repository URL (public / access granted to judges)	https://github.com/dylanchong0820/SDC-Hackathon.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

QuoteBot AI operates through a zero-friction, end-to-end architecture starting at the user interface, where SME users send unstructured text, voice notes, quote images, or PDFs via WhatsApp. Meta's WhatsApp Cloud API captures these incoming messages and forwards webhooks to a cloud-hosted FastAPI backend microservice, which immediately routes heavy tasks into an asynchronous Redis job queue. Background workers then dispatch the unstructured inputs to multi-modal Vision-LLMs running on serverless RunPod GPU endpoints, where strict JSON Schema constraints extract and validate key vendor, pricing, and line-item data. Once validated, the backend sends the structured JSON to a Puppeteer HTML-to-PDF rendering engine to instantly generate a branded, downloadable quotation report, while simultaneously storing the audit-ready transaction record in a PostgreSQL database with Row-Level Security. Finally, the system dispatches the formatted text summary and downloadable PDF report link back to the user directly inside the WhatsApp chat window.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Phyton	backend, message processing, document handling and AI integration.	
Javascript	browser interactions and comparison progress updates.	
CSS HTML5	website structure, styling and responsive layouts.	
SQL	storing and searching conversations and quotation records.	
flask		

Name & Version	Purpose in Project	License
• **vLLM** — external AI inference server; **Apache-2.0**. • **Qwen2.5-VL-3B-Instruct**, by Alibaba Cloud — pretrained model for text and quotation-image analysis.	Determine word and image	Qwen2.5-VL-3B-Instruct/LICENSE

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Runpod	Link to quotebot Ai	Sponser provided
Whatsapp	Create a whatsapp desk	free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
SQL	Save data for AI generate's result	Develop more function for project

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Deepseek v4.1 flash 1017	Open source	pretrained	MIT License

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Company set	handmade	private	-	yes
Test company	handmade	private	-	yes

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☑ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Runpod	API	Link the llm to let AI able to read the image and word to give a response
Replit	Platform	Develop the Quotebot Ai project

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Core functionality for the prototype, replit is very easy to handle and the credit is very enough for all team members. Almost every function is suggested and develop by replit. Runpod is use to link llm, let Ai able to read and response.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

QuoteBot AI collects basic WhatsApp metadata, including phone numbers and sender names, alongside user-uploaded quotation inputs such as chat text, voice notes, images, and PDFs. Unstructured inputs are transmitted via encrypted HTTPS webhooks to multi-modal Vision-LLMs hosted on RunPod serverless endpoints strictly for structured JSON data extraction, ensuring that no data is retained or used for third-party model training. All sensitive information is encrypted in transit using TLS 1.3 and at rest using AES-256, with extracted transaction records stored in a PostgreSQL database protected by Row-Level Security to maintain strict multi-tenant isolation. Raw uploaded media and temporary PDF files are automatically purged after 30 days, while structured transaction logs are retained for 12 months for SME accounting exports and audit compliance, or erased immediately upon user request.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

QuoteBot AI's primary AI limitations include LLM hallucination during field extraction, OCR inaccuracies on degraded handwritten documents, rate limits from messaging webhooks, and dependency on specialized model infrastructure. To eliminate hallucinations and schema mismatches, the system enforces strict JSON Schema constraints alongside automated mathematical validation checks—such as verifying that line items sum correctly to the total—and prompts users inline if confidence thresholds are missed. Request spikes and rate limits are managed via an asynchronous Redis job queue to prevent API throttling during peak hours. Furthermore, vendor lock-in is mitigated by an abstracted model routing layer that enables seamless fallback between RunPod serverless endpoints, self-hosted vLLM containers, and external multi-modal APIs.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Chin]

[25 October 2026]